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Project Proposal
Smart Power Quality Analyzer

EN2160 - Electronic Design Realization

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1 Problem Identification and Justification

Power quality disturbances, such as voltage fluctuations, harmonic distortion, and poor power factor, are common in electrical grids, particularly in developing countries like Sri Lanka. These disturbances can cause equipment malfunction, reduced efficiency, and increased maintenance costs in industrial and institutional settings.

Evidence of this problem is seen in frequent downtime of manufacturing equipment and sensitive electronic devices in laboratories and hospitals due to voltage sags and transients. Small and medium enterprises (SMEs) and educational institutions often lack access to continuous power quality monitoring solutions because commercial analyzers cost between **LKR 350,000–1,750,000** per unit.

Although commercial analyzers exist, they are expensive and complex, leaving a gap for a low-cost, locally deployable solution.

2 Existing Solutions and Differentiation

Existing power quality analyzers are high-accuracy industrial instruments. Their limitations for widespread adoption include:

- High unit cost (**LKR 350,000–1,750,000**)
- Complexity in setup and configuration
- Limited real-time remote monitoring options

The Smart Power Quality Analyzer differentiates itself by:

- Significantly lower unit cost (approx. **LKR 15,000–25,000**) using low-cost microcontrollers and sensors
- Compact, modular PCB-based design with locally sourced components
- Optional cloud-based monitoring with subscription model for remote access and analytics
- Continuous real-time monitoring with local display

3 Proposed Solution and Technical Implementation

The proposed system is a microcontroller-based embedded device designed for continuous monitoring of power quality parameters.

3.1 Technical Overview

- **Microcontroller:** ESP32 for ADC, signal processing, and Wi-Fi connectivity
- **Voltage Sensor:** ZMPT101B AC voltage transformer module
- **Current Sensor:** SCT-013 split-core current transformer
- **Signal Processing:** RMS voltage/current, real/apparent power, power factor, FFT-based harmonic analysis, Total Harmonic Distortion (THD)
- **User Interface:** OLED or 16x2 LCD display for real-time measurements
- **Data Logging:** MicroSD card for offline storage; optional cloud integration
- **Power Supply:** Isolated AC-DC or buck converter module

The device samples voltage and current signals via conditioned ADC inputs, calculates power parameters in real time, and performs FFT to identify harmonic content. Abnormal conditions like voltage sags or high harmonics trigger alerts both locally and via the cloud.

3.2 PCB and Component Considerations

Design follows EDR best practices:

- Two-layer PCB layout with careful separation of analog and digital signals
- Component selection based on voltage/current ratings, tolerance, and temperature stability
- Decoupling and filtering for ADC inputs to minimize noise
- Enclosure designed for electrical isolation, durability, and user safety
- Connectors and terminals chosen for secure connections and ease of maintenance

Potential failure modes, including overvoltage, sensor drift, and thermal stress, are mitigated through protective circuitry, calibration, and PCB layout optimization.